GitHub Copilot for data scientists

Less boilerplate.
More time for scientific judgment.

Complete code. Ask better questions.
Delegate bounded work, then verify.

A practical briefing for Desjardins teams

ONE WORKSPACE

01 **Explore**
Understand data and assumptions

02 **Experiment**
Build and compare reproducible baselines

03 **Validate**
Test code, metrics and leakage risks

Three ways to work with Copilot

Choose the interaction by how much work you want to delegate.

01 / COMPLETION

Keep typing

Inline suggestions

Finish a pandas transform, a SQL query or a test assertion.

You steer every edit.

```
# aggregate transactions
# by customer and month
```

02 / ASK

Think it through

Conversational help

Explain a notebook, compare metrics or challenge a split.

Advice first; apply deliberately.

```
"Could this feature leak
future information?"
```

03 / AGENTIC

Delegate a task

Agent + tools

Edit related files, run tests and iterate on a bounded task.

Review diffs and tool actions.

```
"Build a baseline pipeline
and run its unit tests."
```

More autonomy does not mean less accountability. Keep context scoped and permissions intentional.

RPI: turn a question into verified work

HVE Core on GitHub: Research -> Plan -> Implement, with review and follow-up.

R Research

Understand before editing

Inspect data contracts and code.

Check leakage and alternatives.

Separate evidence from assumptions.

Output: findings + open decisions

`/rpi-research`

P Plan

Make success testable

Choose the split and metrics.

Sequence small, bounded tasks.

Define tests and acceptance criteria.

Output: plan + critique

`/rpi-plan`

I Implement

Execute the approved scope

Build the pipeline and tests.

Run checks; record real results.

Review diffs against the plan.

Output: changes + validation

`/rpi-implement`

Reuse adequate research | Review the plan | Validate results | Route gaps back to the right phase

Right-size the model to the uncertainty

Start with routing, or choose capability manually. Escalate when evidence says you should.

AUTOMATIC ROUTING

Auto / Hydrafusion*

Let the available router balance

capability, latency and cost.

Use approved options only.

Check the picker and admin policy.

MANUAL CAPABILITY CHOICE

LARGE	**Research + rubber ducking** Challenge assumptions, compare approaches, look for leakage and counterexamples.
MEDIUM	**Planning** Turn evidence into tasks, dependencies and measurable acceptance criteria.
SMALL / MEDIUM	**Implementation** Small for routine edits; medium for complex changes. Escalate if failures or ambiguity persist.

*Hydrafusion is setup-dependent, not a universal Copilot option. Tiers above are a practical heuristic, not product labels.

Try it: a reproducible credit-risk baseline

One synthetic dataset. One clear objective. Three bounded prompts.

01

RESEARCH / ASK

"Inspect this synthetic dataset and notebook.
Identify leakage risks and compare validation splits.
Do not edit files yet."

02

PLAN / HVE RPI

"Plan a reproducible baseline with a time-based split.
Fit preprocessing on training data only.
Define metrics and tests before coding."

03

IMPLEMENT / AGENT

"Implement the approved plan and run the tests.
Report PR-AUC, calibration and subgroup results.
Show the diff and any remaining limitations."

DONE MEANS

A clean rerun works.
Split and leakage tests pass.
Metrics include limitations.
A human reviews the result.

Faster work. Same rigor.

Banking guardrail: synthetic or approved de-identified data only; no client PII or secrets in prompts.